

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	Not Provided
Project Name	Smart Lender
Maximum Marks	5 Marks

Functional Features Overview:

This document lists the main functional features implemented in Smart Lender. The project combines machine learning, web interaction, batch processing, session tracking, and reporting to support loan–screening use cases.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Single Applicant Loan Prediction	Accepts applicant financial and demographic inputs and predicts loan approval or rejection.	app.py , templates/index.html , templates/result.html	Done	High

2	JSON Prediction API	Exposes machine – learning inference through <code>/api/predict</code> for programmatic integration.	<code>app.py</code>	Done	High
3	EMI Calculator API	Calculates EMI, total payment, interest, and DTI using <code>/api/emi</code> .	<code>app.py</code>	Done	Medium
4	Batch CSV Upload and Scoring	Allows uploading CSV files to predict multiple applications at once.	<code>app.py</code> , <code>templates/batch.html</code>	Done	High
5	Batch Result Download	Generates downloadable CSV output for scored batch uploads.	<code>app.py</code>	Done	Medium
6	Session–Based Prediction History	Stores recent prediction records within the active user session and displays them on the history page.	<code>app.py</code> , <code>templates/history.html</code>	Done	Medium

7	Model Analytics and Reports	Displays model comparison information, feature importances, and generated reports/charts.	app.py , templates/about.html , templates/reports.html , static/reports	Done	Medium
8	Model Training and Selection Pipeline	Cleans data, engineers features, trains multiple classifiers, selects the best model, and stores artifacts for deployment.	train_model.py, model/	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Real-time loan prediction, model training pipeline, batch CSV scoring, JSON API access, and session history tracking

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	4	Home application form, result page, batch upload page, history/report views

2	Backend / Logic	3	Prediction workflow, preprocessing pipeline, EMI computation
3	Database / Storage	2	Model artifact storage in model/, temporary session storage for history and batch CSV output
4	API / Integration	3	/api/predict, /api/emi, /health
5	Security / Authentication	1	Basic Flask session secret and input validation; no dedicated login or authentication module is implemented